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CYCLIC VARIATION OF POTASSIUM CONDUCTANCE IN A BURST-GENERATING NEURONE IN *APLYSIA*

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SUMMARY

1. The hyperpolarization between bursts in the R15 cell of *Aplysia* is accompanied by an increase in membrane slope conductance.

2. The post-burst hyperpolarization can be observed with ouabain, lithium, or potassium-free solution if artificial inward current is applied. The hyperpolarization can be observed with dinitrophenol or cooling to 10° C, with no injected current. Thus, the hyperpolarization apparently is not due to the cyclic activity of an electrogenic pump.

3. A reversal potential for the post-burst hyperpolarization can be demonstrated by passage of inward current during the inter-burst period. The reversal of direction of the potential depends on recent occurrence of a burst.

4. The reversal potential varies with external potassium concentration, but not with chloride or sodium.

5. The post-burst hyperpolarization is not blocked by external tetraethylammonium at a concentration which greatly prolongs the action potentials.

6. During the onset of spike blockage by, and recovery from, calcium-free + tetrodotoxin saline, the bursts of action potentials appear to be driven by endogenous waves of membrane potential.

7. The hyperpolarizing phase of the waves in calcium-free + tetrodotoxin medium is accompanied by an increased slope conductance.

8. A reversal potential can be demonstrated for the hyperpolarization following a wave in calcium-free + tetrodotoxin medium by applying inward current during the interwave period.

9. The waves in calcium-free + tetrodotoxin medium are blocked by ouabain but can be reinstated by artificial hyperpolarization.

10. The post-burst hyperpolarization and the post-wave hyperpolarization appear to result from a periodic increase in membrane conductance, primarily to potassium ions.

INTRODUCTION

The neurone R 15 (Frazier, Kandel, Kupfermann, Waziri & Coggeshall, 1967) in the abdominal ganglion of *Aplysia californica* has an endogenous bursting pattern (Alving, 1968; Frazier *et al.* 1967; Strumwasser, 1967) with a circadian rhythm that corresponds to the animal's behavioural activity (Strumwasser, 1965). The function of R 15 is not as yet fully understood; the whitish colour of the cell and the presence of granules with dense cores which develop during the life of the animal suggest a possible neurosecretory function (Frazier *et al.* 1967). The activity of R 15 is inhibited by the application of hypotonic saline or seaweed extract to the osphradium (Jahan-Parwar, Smith & von Baumgarten, 1969; Stinnakre & Tauc, 1969). Also, direct electrical stimulation of the branchial nerve (which goes to the osphradium) inhibits R 15 (Stinnakre & Tauc, 1966; Frazier *et al.* 1967).

The size (200–500 μm) and durability of R 15 make it an ideal model in which to study the mechanisms responsible for endogenous burst activity in neurones. After a burst in R 15, the membrane is suddenly hyperpolarized (*ca.* 20 mV); the hyperpolarization gradually relaxes and the membrane potential returns to threshold, eliciting another burst. The studies reported here will deal primarily with the mechanisms responsible for the post-burst hyperpolarization.

Hyperpolarization following a burst of action potentials has been attributed to an increased potassium conductance in some preparations (Meves, 1961, in frog single node; Gage & Hubbard, 1966, in rat phrenic terminals; Brodwick & Junge, 1972, in *Aplysia* giant neurone) and to the activation of an electrogenic sodium pump in others (Connelly, 1959, in rabbit vagus C-fibres; Straub, 1961, in frog A-fibres; Holmes, 1962, in rabbit, rat and cat C-fibres; Nakajima & Takahashi, 1966, in crayfish stretch receptors; Rang & Ritchie, 1968, in rabbit vagus C-fibres; Baylor & Nicholls, 1969, in leech ganglion cells; Kuno, Miyahara & Weakly, 1970, in cat dorsal spinocerebellar tract neurones; Sokolove & Cooke, 1971, in crayfish stretch receptors).

Strumwasser (1968, 1971) has proposed that the hyperpolarization in R 15 following a burst is caused by an electrogenic sodium pump which is coupled to transport of chloride. The electrogenic-pump theory has also been supported by the work of Romey & Arvanitaki-Chalazonitis (1970). On the other hand, Carpenter & Alving (1968) have shown that bursting cells in the abdominal ganglion of *Aplysia* show a normal discharge pattern when the sodium pump is pharmacologically blocked and the membrane is hyperpolarized by artificial inward current. More recently, Carpenter (1973) has found the same result in the R 15 cell. In an identified snail

neurone which has characteristics in common with R 15 (bursting pattern, dense granules, circadian rhythm, possible control of osmolarity), Gainer (1972) has convincing evidence that the post-burst hyperpolarization is caused by an increased potassium conductance and not an electrogenic sodium pump.

The results reported here will demonstrate that the hyperpolarizing wave seen in R 15 does not depend upon the cyclic activity of an electrogenic pump, and that during the post-burst hyperpolarization there is an increased potassium conductance.

METHODS

All experiments were performed on the neurone R15 (Frazier *et al.* 1967) in the abdominal ganglion of *Aplysia californica*. The animals were used within a month of capture, as experiments with animals kept beyond this time were usually unsuccessful. The abdominal ganglion was removed and pinned onto a layer of resin (Sylgard 182 encapsulating resin, Dow Chemical) which lined the bottom of the recording chamber. Neurones were exposed by using very fine forceps and iridectomy scissors to dissect away the connective tissue above the cell.

TABLE 1. Composition of solutions (mM)

	Na	K	Ca	Mg	Cl	SO_4	$MeSO_4$	Tris
Normal saline	494	11	11	49	572	30	—	10
K-free	505	—	11	49	572	30	—	10
High-K	472	33	11	49	572	30	—	10
Ca-free	494	11	—	49	562	30	—	26
Low Cl	494	11	11	49	78	30	494	10
Low Na	124	11	11	49	493	30	—	408

Micro-electrodes used for intracellular stimulation and recording were filled with 3 M-KCl or 3 M-K acetate; both gave similar results. The stimulating electrodes had a resistance of 2–10 MΩ; the recording electrodes 5–20 MΩ. The recording electrode and reference electrode (a sea-water-Ag-AgCl agar bridge) were connected to high impedance preamplifiers (ELSA-3, Electronics for Life Sciences) and the difference in preamplifier outputs was displayed on an oscilloscope. The output from the oscilloscope was recorded on a Hewlett-Packard (Model 322) thermal-writing recorder, which had the lowest frequency response in the arrangement. The rise time of the entire circuit including the writing pens was about 10 msec, which was sufficiently fast that action potential overshoots were attenuated less than 10 mV.

The stimulus was delivered through a 60 or 100 MΩ series resistor to the stimulating electrode. During stimulation current was monitored across a 100 kΩ resistor between a chlorided silver wire in the bath and ground. The current was measured on a second channel of the recorder.

The composition of the solutions used is shown in Table 1. The Tris [tris (hydroxymethyl) aminomethane] was neutralized to pH 7·7 with HCl. Tetrodotoxin was applied from a stock solution of 10^{-3} g/ml. diluted 40 times with the solution used, to give a final concentration of 25 μg/ml. Tetraethylammonium was applied by replacing an equivalent amount of sodium in the normal saline. DNP solutions were

acidified to pH 6·9, as the inhibitor reportedly becomes inactive at higher pH (Nakajima & Takahashi, 1966). Solution changes were performed by continuous perfusion of the chamber (volume $\approx$ 3 ml.) until 100 ml. of the new solution had passed through. This method was shown by dye-dilution experiments to result in a thorough exchange of solutions. Cooling was accomplished with a Cambion 806-3000-01 microscope stage cooler with built-in heat sink. Cooling time to 10° C from room temperature (23° C) was about 15 min. All experiments not involving tests of cooling were conducted at 21–23° C.

RESULTS

Potential and conductance in the interburst period. An example of the normal burst pattern of firing of the R 15 cell is shown in Fig. 1. The top lines in each part of this Figure show transmembrane potential, and the bottom lines injected current (action potential peaks were obscured by the oscillograph). In part *A*, no current was applied, and a large (27–30 mV) hyperpolarization was apparent following each burst. The hyperpolarization was measured from the top of the last partial depolarization at the end of the burst to the most negative level reached between bursts. The size of the hyperpolarization, and the interburst interval, usually increased during the course of an experiment. In part *B*, brief (2 sec) inward-current pulses of constant amplitude were applied repetitively during the interburst period. In this way, membrane slope conductance could be measured as $\Delta I/\Delta E$, where ΔI was the inward current applied, and ΔE the resulting change in potential. It can be seen that the slope conductance of this cell was greatest immediately following the burst (about $7{\cdot}2 \times 10^{-7}$ mho), and declined to a minimum value (about $5{\cdot}4 \times 10^{-7}$ mho) just before the start of the next burst. This result suggested a time-dependent change of conductance. However, as shown in part *C*, the membrane slope conductance was also potential-dependent: a constant-amplitude pulse of current was applied at a fixed time after the burst, starting at the spontaneously occurring potential (first and third pulses) and at an artificially hyperpolarized level (second pulse). The size of the potential change due to the pulse depended strongly on the potential level at which the pulse was applied. Hence, it was possible that the conductance change seen in part *B* could have resulted from or been strongly modified by, the potential-dependency of the membrane conductance. Further evidence for a time-dependent conductance change in the R 15 cell is supplied by the experiments on reversal potentials described later.

Effects of procedures which inhibit metabolic pumps. Since the primary hyperpolarization between bursts could have arisen from the cyclic action of an electrogenic pump, it was important to examine the effects of agents which inhibit such pumps. Electrogenic pumps which exert a strong tonic influence on membrane potential have been demonstrated in a variety of

Aplysia neurones (Carpenter & Alving, 1968; Marchiafava, 1970; Brodwick & Junge, 1972), including R 15 (Carpenter, 1970, 1973). Thus, we had to consider whether periodic changes in such a pump could contribute to the hyperpolarization between bursts. To do this, we examined the effects of ouabain, DNP, lithium–saline, potassium-free saline, and cooling, all of which are known to interfere with metabolically driven pumps.

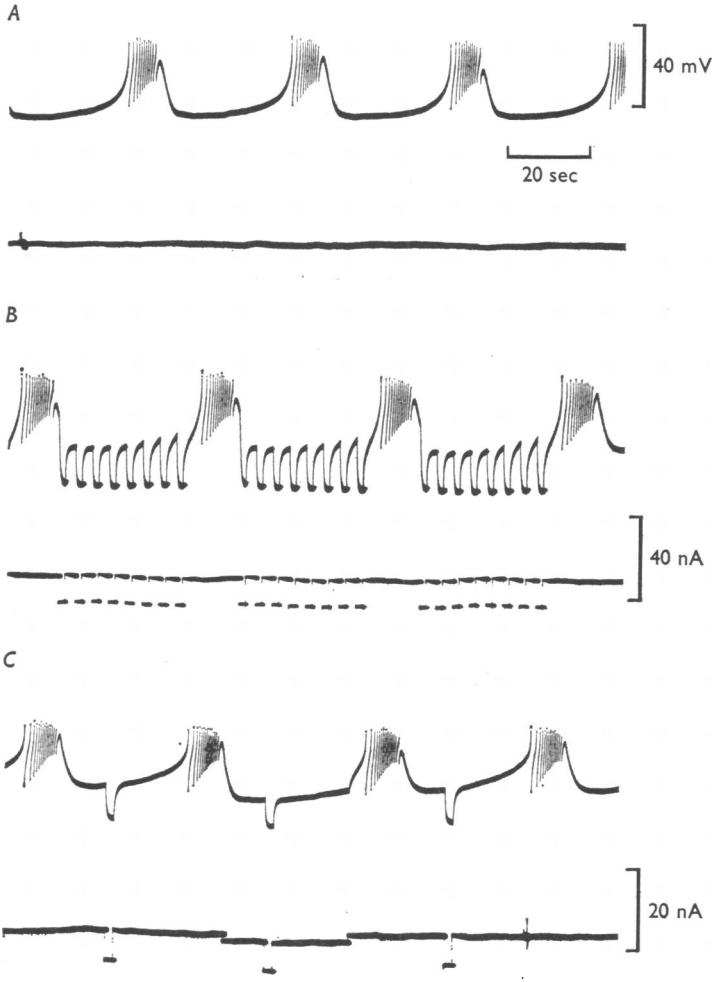


Fig. 1. Potential and conductance changes during interbust period. Top traces in each part show potential, lower traces current. Action potentials obscured by oscillograph. Part *A*: normal bursting with no applied current. Part *B*: regular 2-sec inward current pulses applied to measure slope conductance ($\Delta I/\Delta E$) between bursts. Part *C*: slope conductance increased by artificial hyperpolarizarion (centre). Voltage calibration applies to *A*, *B* and *C*. Current calibration same in *A* and *B*. Cell 12–5.

Ouabain apparently blocks electrogenic pumps by inhibiting the sodium–potassium-dependent ATP-ase in the membrane (Den Hertog & Ritchie, 1969), and eliminates pump-dependent hyperpolarizations in a variety of preparations (Carpenter & Alving, 1968; Baylor & Nicholls, 1969; Gorman & Marmor, 1970*a*; Livengood & Kusano, 1972; Koike, Brown & Hagiwara, 1971). The result of application of 0·4 mM ouabain to the R 15 cell is shown in Fig. 2. Upon introduction of ouabain-containing saline into the chamber

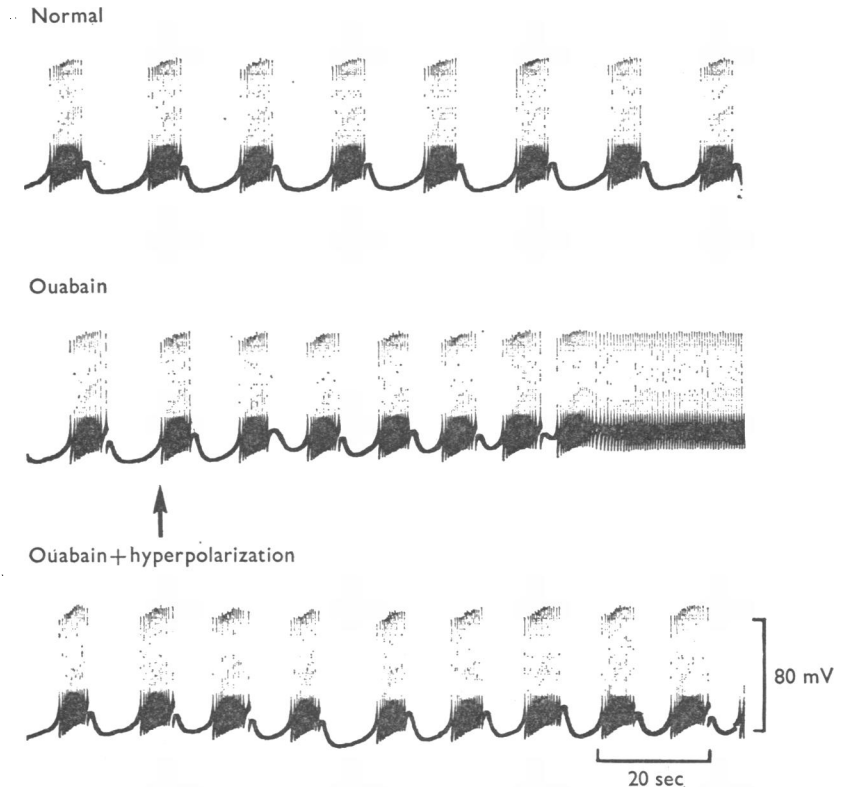


Fig. 2. Effect of ouabain. At the arrow in the middle trace, saline containing 0·4 mM ouabain admitted to chamber. Bottom trace shows restoration of bursting by artificial hyperpolarization. Cell 11–6.

(arrow), the hyperpolarization between bursts was reduced and the interburst interval was shortened. After about a minute, the cell began to discharge regularly. At this time, by passing a maintained inward (hyperpolarizing) current through the membrane, it was possible to reinstate the burst pattern of firing and to restore both the hyperpolarization and the interburst interval to normal values. This effect could be demonstrated for about 15 min in the presence of ouabain; with longer applications the cells

became incapable of producing any bursts. The regular discharge seen with ouabain could also be produced by applying outward current to the cell in normal saline. Treatment with ouabain evidently had a deleterious effect on the cells, as reflected by our failure to record normal bursting patterns after washing out the chamber with normal saline. A similar irreversibility of the effects of ouabain has been noted in other types of cells (Carpenter & Alving, 1968; Baylor & Nicholls, 1969; Romey & Arvanitaki-Chalazonitis, 1970; Koike *et al.* 1971; Junge, 1972). However, the result of treatment with ouabain for up to 15 min was consistent with blockage of an electrogenic pump, causing depolarization of the cell. That we were able to observe normal bursting in the presence of ouabain with artificial inward current suggested that an electrogenic pump mechanism could not account for the hyperpolarization between bursts.

DNP, by interfering with the production of ATP, can also block electrogenic pumps in excitable membranes (Nakajima & Takahashi, 1966; Airapetyan, 1969; Godfraind, Kawamura, Krnjević & Pumain, 1971; Rheuben, 1972). In one R 15 cell, application of 0·3 mM-DNP in normal saline at pH 6·9 caused a depolarization of about 10 mV within 4 min, but did not block bursting. After 16 min in DNP the bursting was finally blocked, but the last burst was followed by a 17 mV hyperpolarization, indicating that the mechanism of the hyperpolarization had not been blocked. The DNP evidently produced irreversible changes in the cell, as no recovery of bursting was ever obtained after treatment with DNP, even with repeated saline washes. A similar harmful effect of DNP has also been noted on the giant (R 2) cell (Brodwick & Junge, 1972). However, during the first 15 min or so in DNP, after the depolarizing action had occurred, the bursting pattern of the R 15 cell continued in a normal-appearing manner.

Lithium ions apparently cannot act as a substrate for sodium pumps, and replacement of external sodium with lithium blocks the electrogenic action of such pumps (Nakajima & Takahashi, 1966; Carpenter & Alving, 1968; Rang & Ritchie, 1968; Livengood & Kusano, 1970; Sokolove & Cooke, 1971; De Weer & Geduldig, 1973). Fig. 3 shows the effect of lithium–saline on the R 15 cell (action potentials obscured by the oscillograph): in normal saline (top line) the hyperpolarization between bursts was about 13 mV, and there were about 13 spikes per burst. About 1 min after introduction of lithium–saline into the chamber (arrow, second line), the interburst interval was increased but no large change in the hyperpolarization was seen. After 12 min in lithium-saline (third line) the hyperpolarization was reduced to 7 mV and there were 11 spikes per burst. By this time the average potential during the interburst period had become about 12 mV more positive than in normal saline. As shown in the fourth

line, application of artificial inward current in the presence of lithium increased the post-burst hyperpolarization to 15 mV and the number of spikes per burst to 16. Even without artificial inward current, this cell continued to discharge in bursts for 25 min in lithium–saline. When normal saline was placed in the chamber, the hyperpolarization recovered to about

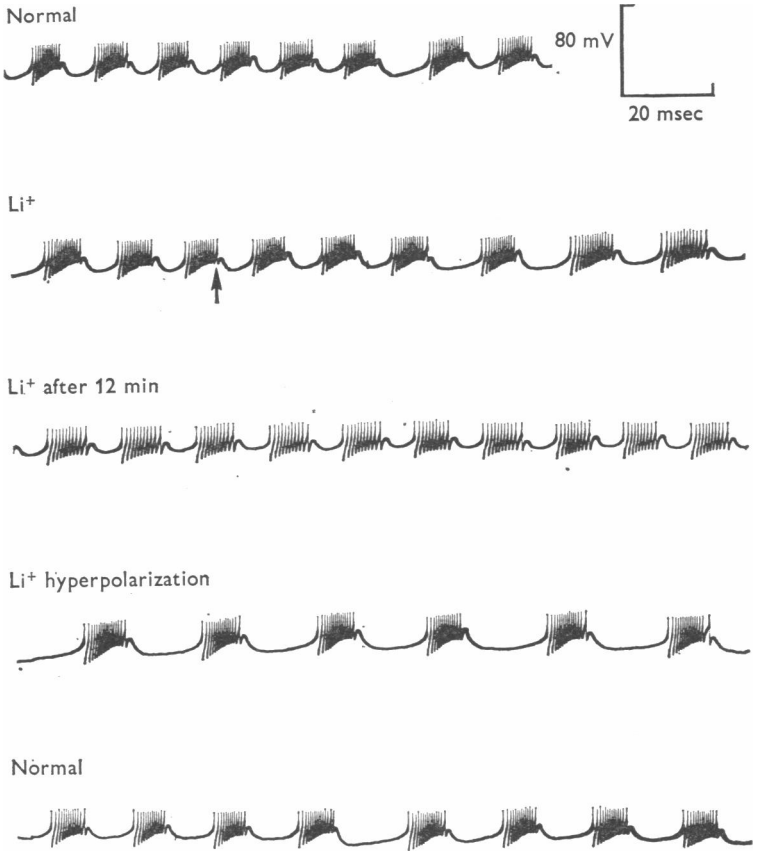


Fig. 3. Effect of replacement of sodium with lithium. Spikes obscured by oscillograph. At arrow in second line, lithium–saline admitted to chamber. After 12 min, number of spikes per burst and post-burst hyperpolarization somewhat reduced. Passage of artificial inward current restored number of spikes and hyperpolarization to normal values. Cell 11–2.

11 mV, and the spike-count per burst became about 13, the same as previously seen in normal saline. Good recovery after treatment with lithium has been found in other preparations (Nakajima & Takahashi, 1966; Rang & Ritchie, 1968; Sokolove & Cooke, 1971). The main effect of lithium on this and other cells studied at 21° C appeared to be a depolariza-

tion which developed over a period of about 10 min, and which could be completely offset by artificial inward current. In a cell examined at 12–15° C, the effects of lithium–saline were slightly more complicated: about 1·5 min after introduction of lithium at 12° C, the cell stopped firing and the membrane potential became constant. About 5·5 min later, regularly occurring hyperpolarizing waves without action potentials were observed, similar to those seen in calcium-free medium containing tetrodotoxin (Strumwasser, 1971). The waves were 20 mV in amplitude, and could be augmented to 35 mV by applying artificial inward current. When the preparation was warmed to 21° C, two or three action potentials occurred on the positive peak of each wave. Upon replacement of normal saline in the chamber, the bursts took on a normal appearance. Although the mechanism of suppression of spikes at 12–15° C by lithium–saline was unclear, it was possible to observe large regular hyperpolarizations for up to 20 min in this solution. This observation indicated that the hyperpolarizations could not have resulted from the action of an electrogenic pump. Since the burst pattern of firing could be reinstated in the presence of lithium by warming to 21° C, it appeared that the principal effect of cooling was to inhibit the production of action potentials.

All outward-directed electrogenic pumps which we know of are partially coupled to potassium influx, and are thus inhibited in potassium-free medium (Connelly, 1959; Kerkut & Thomas, 1965; Nakajima & Takahashi, 1966; Rang & Ritchie, 1968; Carpenter & Alving, 1968; Thomas, 1969; Koike *et al.* 1971; Sokolove & Cooke, 1971; Livengood & Kusano, 1972). In our experiments, replacement of external potassium with sodium never blocked the burst pattern of firing. In one cell, the post-burst hyperpolarization in normal saline was 16 mV and the interburst interval was about 10 sec. Introduction of potassium-free saline into the chamber caused a depolarization of the interburst potential of about 20 mV after 3 min, and reduced the post-burst hyperpolarization to 6 mV and the interval to about 4 sec. Application of artificial current restored the hyperpolarization to 14 mV and the interburst interval to 8 sec. This result suggested that potassium-free medium simply inhibited the tonic electrogenic pump, causing depolarization. Applications of potassium-free solutions to other cells for up to 45 min also failed to block the burst pattern of firing, even with no artificial inward current.

Lowering the bath temperature to 5–10° C strongly inhibits electrogenic pumps in neurones (Carpenter & Alving, 1968; Den Hertog & Ritchie, 1969; Baylor & Nicholls, 1969; Gorman & Marmor, 1970*a*; Sokolove & Cooke, 1971; Livengood & Kusano, 1972), and also decreases the resting sodium/potassium permeability ratio (Marchiafava, 1970; Gorman & Marmor, 1970*b*). This permeability change tends to hyperpolarize the cell,

and could not explain a depolarization produced by cooling. The effect of cooling the R 15 cell from 20 to 10° C is shown in Fig. 4. The upper traces in each part show transmembrane potential, and the lower traces bath temperature near the ganglion. In part *A*, at 20° C, the hyperpolarization between bursts was about 20 mV, and there were 14–15 spikes per burst. Cooling to 10° C (part *B*), which took about 15 min, reduced the hyper-

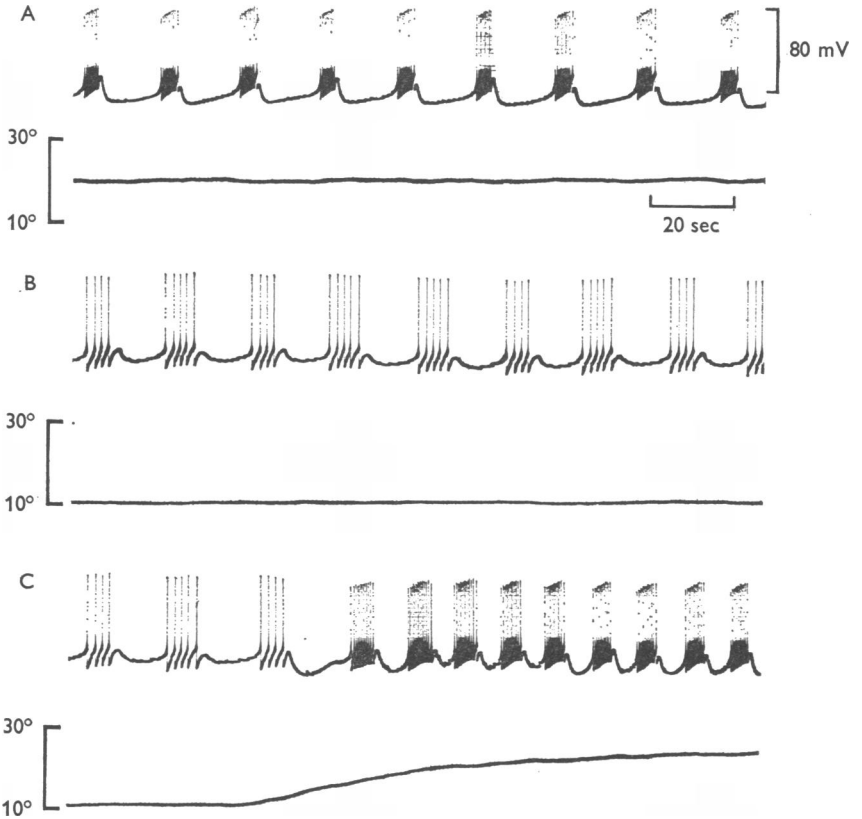


Fig. 4. Effect of cooling and rewarming. Top traces in each part show potential, lower traces bath temperature near ganglion. Part *A*: normal bursts at 20° C. Part *B*: bursts after cooling to 10° C. Part *C*: warm saline introduced into chamber, raising temperature to 24° C. Cell 11–14.

polarization to 10 mV and the spike count to 4–5. At this time, the average potential in the interburst period was about 20 mV more positive than at 20° C. The cell was able to maintain the burst pattern of firing at 10–11° C for 45 min. Upon sudden introduction of warm saline into the bath (part *C*), the temperature increased to 24° C, the hyperpolarization to about 18 mV, and the spike count to 15–16 per burst. The average potential

during the interburst period also returned to a more negative level. After the warm saline had been in the chamber for 10 min, it reached a temperature of 20° C and the interburst interval lengthened to that seen in part *A*. The principal effect of cooling was a depolarization of the cell, suggesting inhibition of a tonic electrogenic pump. The persistence of the post-burst hyperpolarization at 10° C suggested that it was not produced by a cyclic electrogenic pump.

All of the pump-inhibiting procedures tested produced a general depolarization during the interburst period and a reduction of the post-burst hyperpolarization. In the cases of ouabain, lithium and potassium-free saline, we were able to observe the normal bursting pattern by passage of artificial inward current. These results indicated that the main effects of the procedures used was to inhibit the tonic electrogenic pump, and depolarize the cells. The fact that we could obtain normal burst patterns under these conditions strongly suggested that the post-burst hyperpolarization was not due to the cyclic action of an electrogenic pump.

Reversal potential for the post-burst hyperpolarization. Since a pump mechanism as considered above could not explain the hyperpolarization following a normal burst, an alternate hypothesis was an increased conductance to some ion or ions with an equilibrium potential more negative than resting, or a decreased conductance to ions with a more positive equilibrium potential. If such a conductance change were occurring, it should then be possible to measure a reversal potential for the post-burst hyperpolarization (Nakajima & Takahashi, 1966; Brodwick & Junge, 1972; Junge, 1972). The measurement of a reversal potential in the R 15 cell is shown in Fig. 5. The upper traces in each part show membrane potential and the lower traces applied current. In part *A*, no current was injected, and the post-burst hyperpolarization was about 16 mV. In part *B* a 30 sec inward-current pulse was applied immediately after the burst; the membrane potential was slightly hyperpolarized by the inward current, but retained a down-then-up time-course. In part *C* the inward current was increased, and the membrane potential became essentially flat during the inward current. In *D* and *E*, the inward current was increased still further, and the potential after the burst showed a continuously downward trend. To describe experiments of this type, we defined E_H, the final holding potential reached during the inward-current pulse, and ΔE, the difference between E_H and the potential immediately after the start of the inward current. The reversal potential for the post-burst hyperpolarization was taken as that value of E_H for which $\Delta E = 0$, about -67 mV in this cell.

The reversal of direction of the post-burst potential with artificial inward current constituted evidence for time-dependent conductance change in the R 15 cell (with a constant applied current, the potential change was

least immediately after the burst and increased with time). However, it was possible that the reversal could have resulted from a direct effect of the strong inward current on the membrane. To examine this question we applied inward currents with and without an immediately preceding burst,

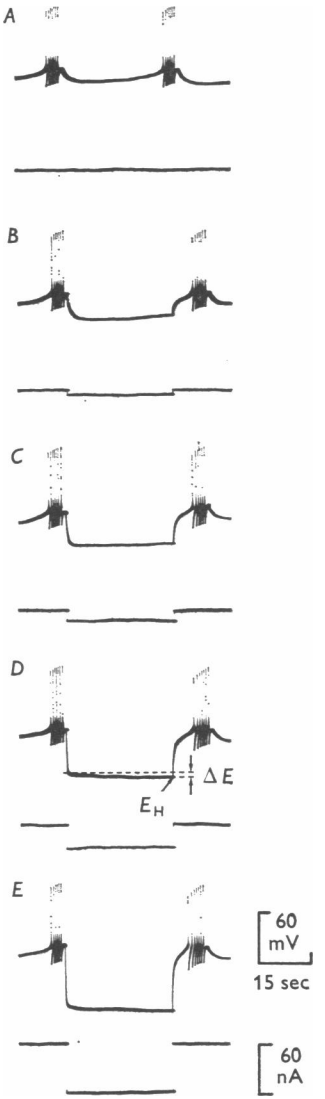


Fig. 5. Reversal potential of the post-burst hyperpolarization. Top traces in each part indicate potential, lower traces current. Part *A*: no current applied. Parts *B–E*: inward-current steps of increasing amplitude applied between bursts. Potential during inward current became almost flat in *C*, reversed direction in *D* and *E*. Reversal potential (-67 mV) taken as value of E_H where $\Delta E = 0$. Cell 10–31.

as shown in Fig. 6. In part *A*, injection of inward current immediately after a burst produced the normal reversal of direction of potential (action potential peaks not shown). However, when the same inward current was applied at the end of the interburst period, just before a burst would have occurred, much less potential change was seen during the inward current.

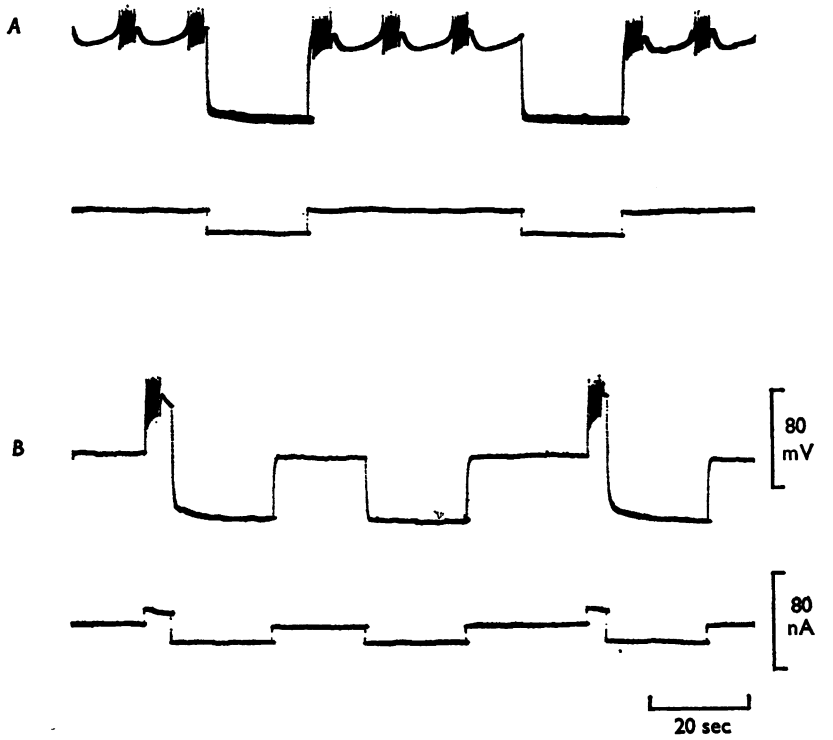


Fig. 6. Importance of preceding burst for reversal. Top traces in each part show potential, lower traces current. Part *A*: inward-current pulse immediately after burst produces reversal of direction of potential. Same current applied at end of an interburst period produces less reversal. Part *B*: maintained inward current applied throughout, to prevent spontaneous bursting. Inward current following an outward-current stimulus produces reversal (first and third inward-current stimuli). Same current with no preceding burst shows less reversal (centre). Cell 2–9.

In part *B*, a constant inward current was applied before the start of the trace, to prevent any spontaneous bursting. Then a burst was produced with a 5 sec outward-current stimulus, and additional inward current was applied immediately after the burst. This procedure gave a large reversal of potential during the inward current, similar to that seen when inward current was applied following a spontaneously occurring burst. In the centre of part *B*, the same inward current was applied with no preceding

burst, and almost no reversal was seen. Following this, the first procedure was repeated with essentially the same result as in the first trial, indicating that no long-term inhibition of the reversal mechanism had occurred. The slight reversal of direction of potential with no preceding burst was apparently caused by the stimulus itself, perhaps by depletion of extracellular potassium in a relatively non-diffusible compartment around the cell (Eaton, 1972). The much larger reversal following a burst suggested that a burst could trigger an increase in membrane conductance to some ion or ions whose equilibrium potential was more positive than the membrane potentials during the inward current. Thus, the membrane potential was initially drawn toward the equilibrium potential, and subsequently became more negative as the conductance declined. The reversal could not have been due to cumulative injection of chloride ions into the cell from the stimulating electrode, as this would tend to shift the potential positively during the stimulus. Moreover, this type of reversal is not seen with electrogenic pump mechanisms (Nakajima & Takahashi, 1966; Rang & Ritchie, 1968; Livengood & Kusano, 1972).

Effect of external potassium concentration. The potassium equilibrium potential for the R 15 cell, E_K, has been calculated from direct measurements of internal and external activities as -75 mV (Kunze, Walker & Brown, 1971). Since this is more negative than any potential attained by the cell during normal bursting, an increase in membrane potassium conductance could produce the post-burst hyperpolarization. One expected result with this mechanism would be a sensitivity of the reversal potential, as measured in Fig. 5, to external potassium concentration (Nakajima & Takahashi, 1966; Brodwick & Junge 1972). This was examined in the experiment shown in Fig. 7, using the cell of Fig. 5. Negative values of ΔE corresponded to the down-then-up direction of potential, and positive values to the reversed direction. In normal saline (potassium 11 mM), the reversal potential was about -67 mV. When external potassium was raised to 33 mM (in exchange for sodium), the reversal potential became -51 mV. When the external potassium was completely replaced with sodium, it was no longer possible to reverse the direction of the potential following onset of the inward current, and the entire curve of ΔE *vs.* E_H was shifted to the left. A good recovery of the curve was seen upon returning to normal external potassium concentration.

The change in reversal potential upon increasing the external potassium concentration three times was 16 mV, which corresponded to 34 mV/ten-fold change in potassium. While this value was less than would be expected if the membrane behaved like a potassium electrode, the result strongly suggested that an increase in potassium conductance could contribute to the post-burst hyperpolarization.

Effect of external chloride concentration. Kunze *et al.* (1971) calculated the chloride equilibrium potential for the R 15 cell from direct activity measurements as −52 mV. This was close to the average membrane potentials during the interburst period which we observed, and we did not expect that a change in chloride conductance would contribute greatly to the normal post-burst hyperpolarization. This question was examined, first by observing the effect of chloride reduction on the bursting rhythm, and then by measuring the effect of this procedure on the reversal potential for the post-burst hyperpolarization.

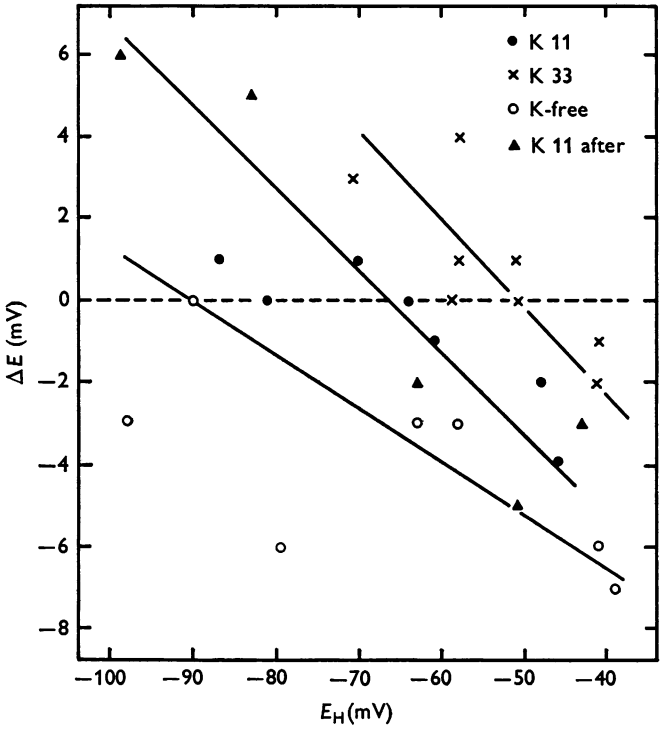


Fig. 7. Effect of external potassium concentration on reversal potential. ΔE and E_H defined as in Fig. 5. Negative values of ΔE correspond to down-then-up direction of potential between bursts. Potassium concentration varied in exchange for sodium. Reversal potential = value of E_H for which $\Delta E = 0$. Cell 10–31.

Three R 15 cells were perfused for up to 30 min with a solution in which all but 1/7·3 of the chloride was replaced with methanesulphonate and all continued to fire in bursts throughout the perfusion. In one cell, 4 min after introduction of low-chloride solution into the chamber, the interburst interval increased from 25 to 35 sec, with no noticeable change in either

the post-burst hyperpolarization or average membrane potential between bursts. Thirty min later the interburst interval was 25 sec and the post-burst hyperpolarization was reduced to about two thirds of the initial value. Complete recovery of the hyperpolarization was seen upon refilling the chamber with normal saline. Although chloride reduction did have some effect on the post-burst hyperpolarization, it did not substantially alter the pattern or timing of the bursting in this cell. In the other cells treated

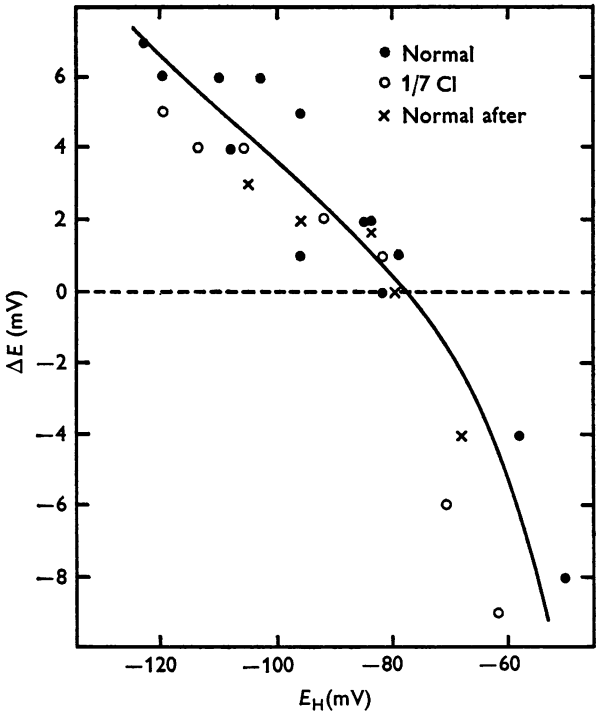


Fig. 8. Small effect of external chloride concentration on reversal potential. ΔE and E_H defined as in Fig. 5. In 1/7 chloride solution, all sodium chloride in saline replaced with sodium-methanesulphonate. Cell 11–22.

with low-chloride solution, the interburst interval was occasionally prolonged up to 1 min, but the bursting was not completely interrupted for up to 30 min in this solution.

The effect of reduction of external chloride concentration on the reversal potential of the post-burst hyperpolarization is shown in Fig. 8. Values of ΔE and E_H were measured as in Fig. 5. The curves obtained in normal saline and low-chloride solution essentially overlapped, indicating little effect of chloride reduction on the reversal potential (although a change of 3–4 mV could have occurred within experimental error). This result

suggested that chloride ions contributed little if any to the increase in membrane conductance following a burst.

Effect of external sodium concentration. A number of studies have indicated that *Aplysia* neurones in general have a high resting sodium conductance (Carpenter & Alving, 1968; Sato, Austin, Yai & Marhuhashi, 1968; Brown, Walker & Sutton, 1970; Carpenter & Gunn, 1970; Marchiafava, 1970). Thus, a decrease in sodium conductance following a burst

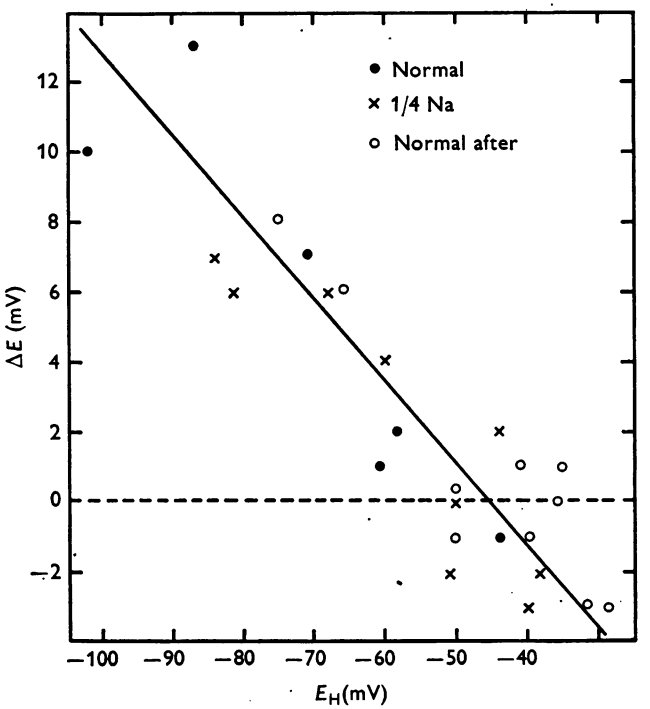


Fig. 9. Small effect of external sodium concentration on reversal potential. ΔE and E_H defined as in Fig. 5. In 1/4 sodium solution, 3/4 sodium chloride in saline replaced with Tris-HCl. Cell 12–12.

could contribute to the post-burst hyperpolarizarion. To test this possibility, we examined the effect of reduction of external sodium concentration on the reversal potential as defined in Fig. 5; the result is shown in Fig. 9. The curves obtained in normal saline, 1/4-normal Na and normal afterward overlapped considerably, although a change in reversal potential of 5 mV could have gone undetected. This result indicated an insensitivity of the reversal potential to external sodium concentration, and suggested that changes in sodium conductance did not contribute to the post-burst hyperpolarization.

With the degree of precision of these experiments, it was not possible to assign relative percentages to the contributions of potassium, chloride, and sodium to the conductance increase. Nevertheless, the results indicated that the major change following a burst was an increased conductance to potassium ions.

Effect of tetraethylammonium ion. Externally applied tetraethylammonium chloride (TEA) interferes with outward potassium currents associated with spike repolarization in the R 2 cell (Junge & Geduldig, 1967) and the R 15 cell (Eaton, 1972). However, external TEA does not block a very slow potassium channel in the R 2 cell which is activated by depolarization (Brodwick, 1972). Thus, TEA may have separate effects on different potassium channels in the same neurone. If the post-burst hyperpolarization in R 15 were blocked by TEA, this would imply that the spike repolarization channel might contribute to the post-burst hyperpolarization.

We initially tried applying 100 mM-TEA to the R 15 cell, replacing an equal amount of sodium chloride in the saline. In one cell, the normal burst consisted of 21 spikes of about 10 msec duration, and the post-burst hyperpolarization was 19 mV. Two min after introduction of TEA-saline into the chamber, the bursting stopped in the most hyperpolarized state of the membrane potential. The final burst consisted of three spikes, the last of which was almost 700 msec in duration, and the post-burst hyperpolarization was 28 mV. After the bursting had stopped, it was always possible to elicit prolonged action potentials in the presence of TEA, with short depolarizing stimuli. After repeated washing with normal saline, the burst pattern resumed with 3 spikes per burst and a hyperpolarization of about 6 mV. Although 100 mM-TEA completely blocked the burst pattern in this cell, the hyperpolarizations just before the block were larger than normal, and the final burst was followed by a large hyperpolarization. These observations suggested that the mechanism responsible for the hyperpolarization had not been blocked by TEA.

Since only a poor recovery was seen after 100 mM-TEA, we tried reducing the concentration to 30 mM. In another cell, the normal burst consisted of 5 spikes of about 10 msec duration and the hyperpolarization after the burst was 13 mV. Two min after application of 30 mM-TEA, the burst contained two spikes, the second of which was 130 msec in duration, and the hyperpolarization was 14 mV. After 5 min in 30 mM-TEA, there was only 1 spike per 'burst', and the hyperpolarization was reduced to 9 mV. Upon washing with normal saline, the spike count recovered to 4 per burst, and the hyperpolarization increased to 12 mV, or essentially the same pattern as was seen before application of TEA. In this cell, 30 mM-TEA did not block either the burst pattern or the post-burst

hyperpolarization. Rather, the initial effect of TEA was to increase the hyperpolarization, at the time at which the spike was greatly prolonged. The opposite initial effects of TEA on the repolarization and post-burst hyperpolarization mechanisms suggested that these might involve functionally different channels.

Waves in calcium-free saline containing tetrodotoxin. When placed in normal or calcium-free saline containing tetrodotoxin the R 15 cell exhibits waves of membrane potential with a similar time course to the normal bursting cycle (Mathieu & Roberge, 1971; Strumwasser, 1965, 1971).

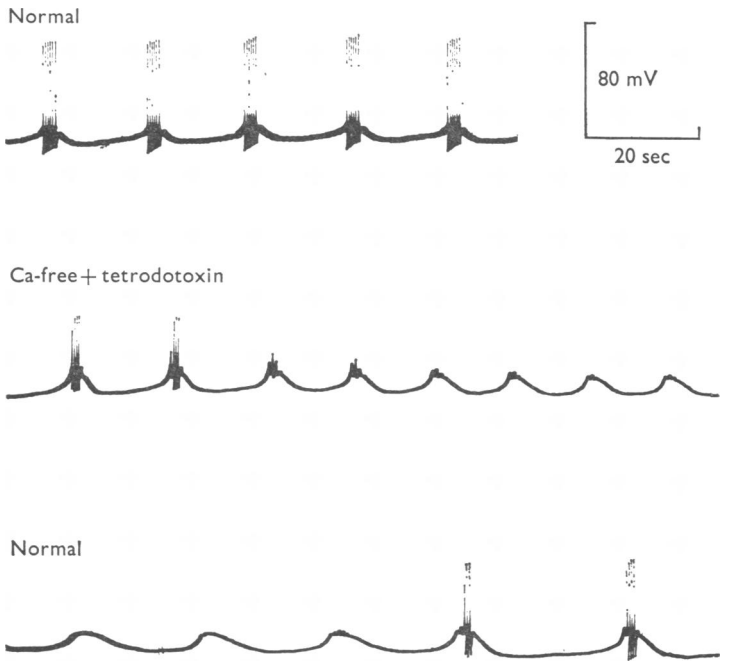


Fig. 10. Relationship of spike activity to waves in calcium-free medium containing tetrodotoxin. Top: normal bursts. Middle: disappearance of spikes in calcium-free medium containing tetrodotoxin 25 μg/c.c., showing persistence of waves. Bottom: recovery of spike activity after washing with normal saline; spikes apparently driven by waves. Cell 11–28.

Sometimes it is necessary to apply a constant inward (hyperpolarizing) current to produce the wave pattern (Mathieu & Roberge, 1971; Strumwasser, 1971). We could reliably observe waves in tetrodotoxin by using calcium-free saline and inward current. Fig. 10 shows the type of wave pattern seen. The top trace is a normal record. Middle trace: failure of spike activity 9 min after the ganglion was perfused with tetrodotoxin 25 μg/ml. in calcium-free saline. The interwave interval was 16 sec, com-

pared to the previous 18 sec interburst interval. In this case, after 5 min the wave pattern stopped and it was necessary to apply a small hyperpolarizing current to continue the wave production. The lower trace illustrates the process of recovery 50 min after the ganglion had been perfused with normal saline; a slight hyperpolarization was applied in this trace. The close similarity of membrane potentials in the interwave and

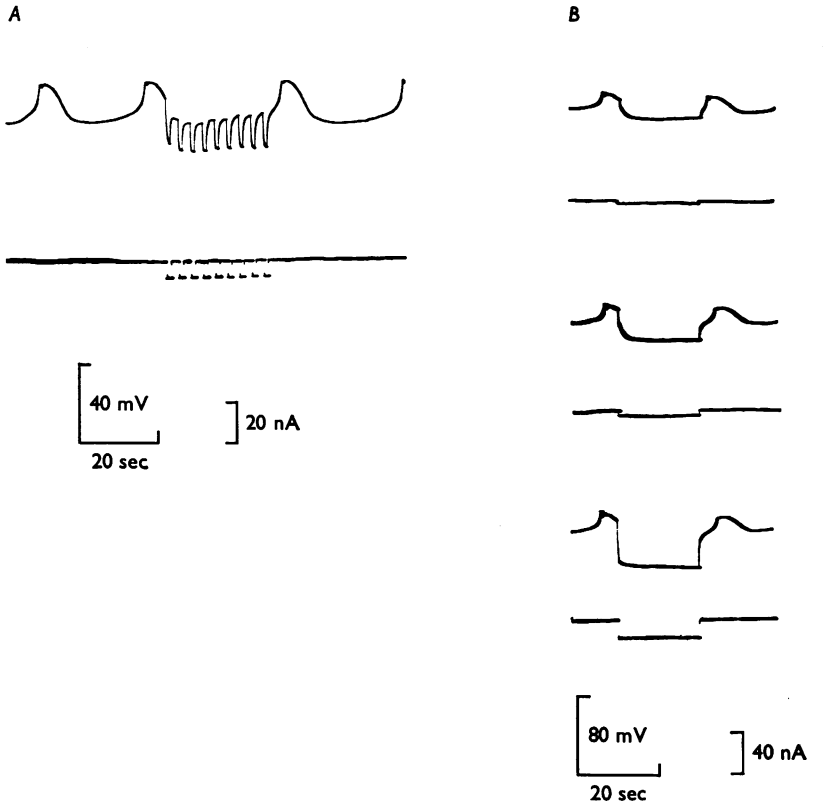


Fig. 11. Conductance change and reversal potential in calcium-free medium containing tetrodotoxin. Part *A*: brief inward-current pulses applied during interwave period. Slope conductance ($\Delta I/\Delta E$) greatest shortly after a depolarizing wave. Part *B*: top traces in each of three sets indicate potential, lower traces current. Application of long inward-current pulses between waves reversed direction of potential (cf. Fig. 5). Cell 11–28.

interburst periods, and the fact that spikes began to 'ride' on the waves as the cell recovered from tetrodotoxin, suggested that the wave mechanism was probably intimately involved in the normal mechanism of burst production.

We wished to see if the hyperpolarization between waves could be

explained by the same type of mechanism (a conductance increase, rather than an electrogenic pump) as the hyperpolarization observed between bursts under normal conditions. Fig. 11, part *A*, shows that the slope conductance varied during the hyperpolarizing phase of the wave, in the same manner as it did during normal bursting (cf. Fig. 1). Further evidence for a conductance increase following a wave was provided by the experiment shown in Fig. 11, part *B*. By applying 20 sec inward-current pulses of increasing amplitude following the wave, it was possible to demonstrate a reversal potential for the post-wave hyperpolarization. The

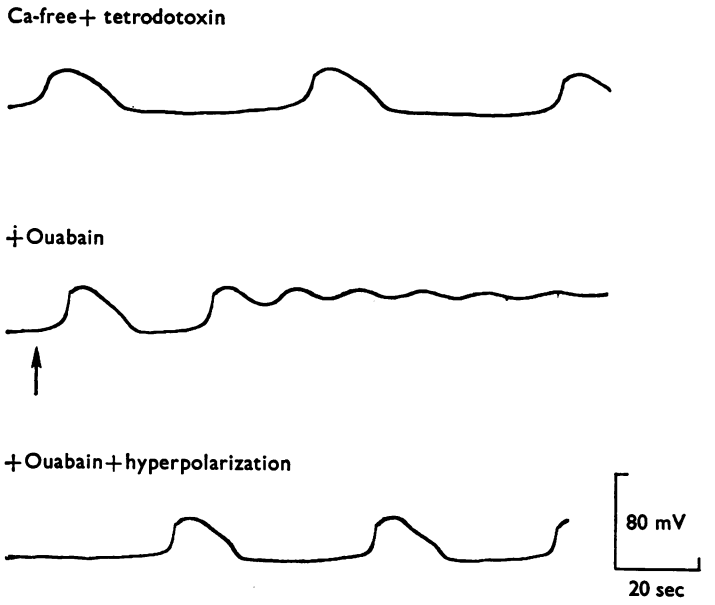


Fig. 12. Failure of ouabain to block waves in calcium-free medium containing tetrodotoxin. Top trace: waves in calcium-free medium containing tetrodotoxin 25 μg/ml., with small maintained hyperpolarization. Middle trace: addition of 0·4 mM ouabain terminated wave pattern. Lower trace: waves restored by additional hyperpolarization. Cell 11-9.

reversal potential for this cell (-70 mV) was similar to that measured for the normal post-burst hyperpolarization (-67 mV in Fig. 5). To examine the possibility that an electrogenic pump could be contributing to the hyperpolarizing phase of the wave, we applied 0·4 mM ouabain to the ganglion in calcium-free solution containing tetrodotoxin. The top trace in Fig. 12 shows the waves in calcium-free + tetrodotoxin 25 μg/ml. with small maintained hyperpolarization. In the middle trace, application of 0·4 mM ouabain in calcium-free + tetrodotoxin solution (arrow) caused a

depolarization and terminated the wave pattern. However, this could be restored by a small increase in the hyperpolarizing current (lower trace). This result suggested that the hyperpolarization following the wave in calcium-free + tetrodotoxin solution was not caused by the cyclic activity of an electrogenic pump.

The slope conductance measured at about the middle of the hyperpolarizing part of the wave in tetrodotoxin calcium-free saline was about twice that measured at a similar point in the normal burst pattern. Although this result indicates that some caution should be used in extrapolating from the wave pattern to the normal burst pattern, our other experiments indicate a strong relationship between the two patterns. The post-wave hyperpolarization and the post-burst hyperpolarization both show an initial conductance increase, a reversal potential for the conductance increase, and a maintained cyclic bursting pattern in the presence of ouabain.

DISCUSSION

The results reported above, and those of studies by other workers (Gainer, 1972; Carpenter & Alving, 1968; Carpenter, 1973), suggest that post-burst hyperpolarization in molluscan neurones is caused by an increase in potassium conductance. This apparently conflicts with studies attributing the hyperpolarization to the activity of an electrogenic pump (Romey & Arvanitaki-Chalazonitis, 1970; Strumwasser, 1968, 1971). One factor which could explain this inconsistency is the apparently harmful effect of ouabain and DNP, when applied for periods longer than about 15 min, after which it becomes impossible to observe bursting even with artificial hyperpolarization. The initial effects of metabolic inhibitors (depolarization, repetitive firing) did not include block of the post-burst hyperpolarization, even though the tonic electrogenic pump had evidently been inhibited. Also, long-term application of some electrogenic pump inhibitors, like ouabain and DNP, to the R 2 cell results in significant intracellular ionic changes (Russell & Brown, 1972). After about 4 hr in ouabain, the intracellular potassium concentration is reduced from 150 to 90 mM (Russell & Brown, 1972). These ionic changes, and the irreversibility of action of ouabain and DNP indicate that only short-term effects should be considered in evaluating the action of the inhibitors. The other pump-inhibiting procedures which we tried (lithium–saline, potassium-free saline, cooling) did not have such irreversible effects, and did not block the hyperpolarization.

Given a steady-state depolarizing drive (such a high resting conductance to sodium), a cyclic increase in potassium conductance could fully account for the bursting activity of the R 15 neurone, as shown schematically in

Fig. 13. It is assumed that the membrane has a relatively high conductance to an ion (probably sodium), with a positive equilibrium potential (E_D), which tends to drive the membrane potential (E_M) of the cell in the depolarizing direction. As the membrane potential becomes more positive, a hyperpolarizing (potassium) conductance mechanism (g_H) is activated after a certain delay, and forces the membrane potential toward the negative equilibrium potential (E_H). The conductance g_H then slowly relaxes, allowing the membrane potential to move positively again. The

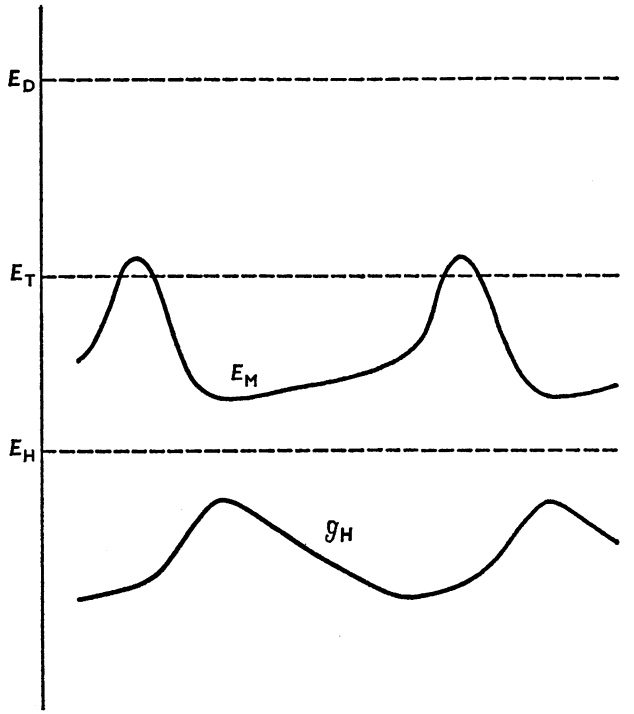


Fig. 13. Model of burst generation, based on cyclic increase of potassium conductance. See text for details.

spikes 'ride' on the depolarizing phase of the cycle which is above the threshold (E_T). An increased frequency of firing at the peak of the depolarizing wave could account for the 'parabolic' curve of intervals *vs.* time during the burst (Strumwasser, 1967). Trautwein & Kassebaum (1961) and Connor & Stevens (1972) have proposed similar models to account for pace-making activity, where only one spike occurs during the depolarizing phase of the cycle.

In addition to a periodic increase in potassium conductance, other

changes are certainly taking place throughout the burst cycle in the R 15 neurone. During the depolarizing phase, there is probably an accumulation of potassium in the infoldings of the cell membrane, shifting the equilibrium potential for the spike repolarization channel in the positive direction (Eaton, 1972; Strumwasser, 1968). This effect cannot explain post-burst hyperpolarization, however, as potassium accumulation shifts the membrane e.m.f. in the opposite direction to the observed hyperpolarization. A possible chloride-dependence of the post-burst hyperpolarization has been suggested by the work of Strumwasser (1968, 1971), in which reduction of external chloride with acetate or propionate abolished bursting. The fact that we could not block bursting with a similar reduction of chloride in exchange for methanesulphonate may reflect a different action of various chloride substitutes on the membrane. For example, the reported recovery after acetate was faster than after propionate (Strumwasser, 1968). Sodium inactivation (Geduldig & Gruener 1970; Hodgkin & Huxley, 1952) would also be expected to occur during the depolarizing phase of the burst cycle, and could affect the termination of the burst. If any decrease in sodium conductance contributed to the post-burst hyperpolarization, it was masked by the increase in potassium conductance, since our experiments on reversal potentials indicated a net increase in membrane conductance following a burst.

Some suggestion as to the mechanism by which depolarization leads to an increase in potassium conductance in the above model is provided by the work of Meech (1972). Injection of calcium into the R 15 cell caused a large increase in potassium conductance. Hence, either influx or internal release of calcium during the depolarizing phase of the burst cycle might be expected to increase the potassium conductance. Depolarization is known to cause calcium entry in the R 15 cell (Stinnakre, Tauc & Saito, 1972), the R 2 cell (Geduldig & Gruener, 1970; Stinnakre, Tauc & Saito, 1972) and in squid axon (Baker, Hodgkin & Ridgway, 1971). Depolarization causes release of internal calcium in muscle (Winegrad, 1968; Ridgeway & Junge, 1973) and in the giant ciliate *Spirostomum* (Ettienne, 1970).

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